Prof. Luiz Paulo Lopes Fávero

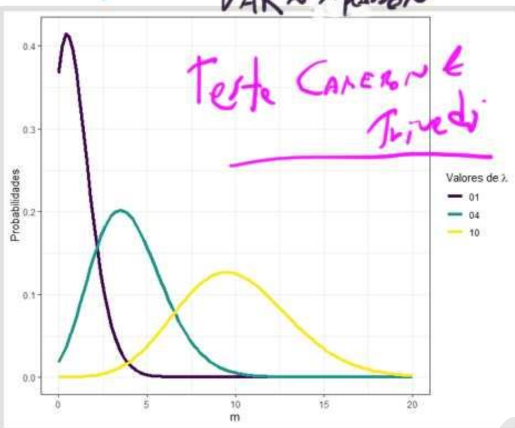
$$p(Y_i = m) = \frac{e^{-\lambda_i} . \lambda_i^m}{m!}$$

Poisson:

$$\ln(\lambda_{poisson_i}) = \alpha + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

$\hat{y}$

$VAR \approx \lambda_{poisson}$




$$p(Y_i = m) = \frac{\delta^{\theta} . m_i^{\theta-1} . e^{-m_i . \delta}}{(\theta - 1)!}$$

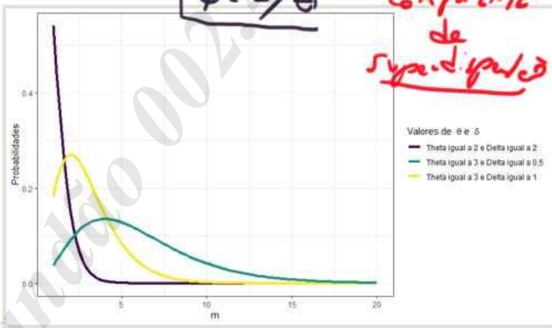
Bin. Neg. II: NB2

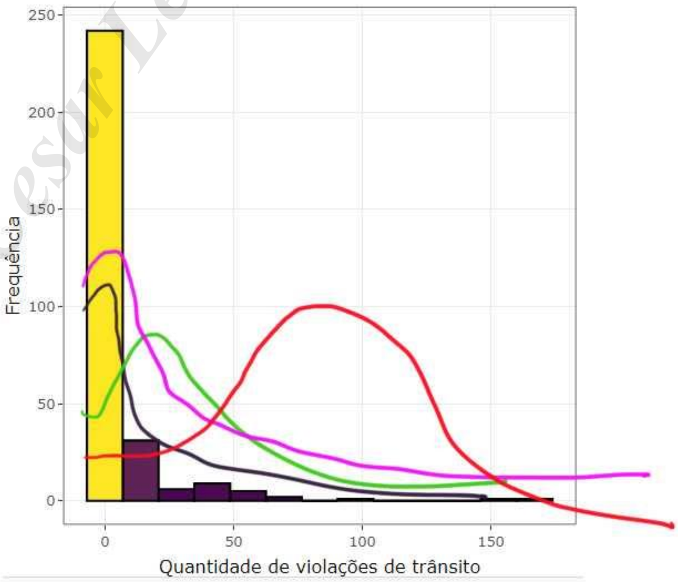
$$\ln(\lambda_{bneg_i}) = \alpha + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

$\hat{y}$

$$VAR = \lambda_{bneg} + \underbrace{\phi(\lambda_{bneg})^2}_{\text{componente de superdispersão}}$$

$\boxed{\phi = 1/\theta}$



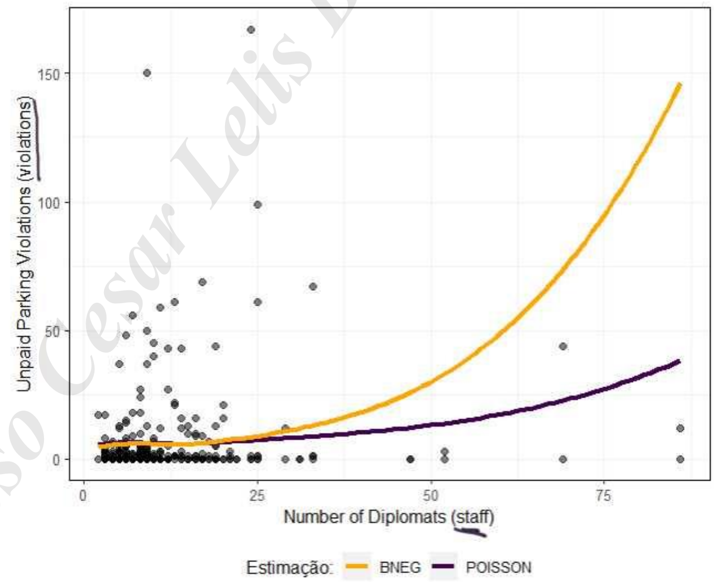
| | Est. | 2.5% | 97.5% | z val. | p |
|---|---|---|---|---|---|
| (Intercept) | 1.9469 | 1.6302 | 2.2636 | 12.0495 | 0.0000 |
| staff | 0.0400 | 0.0226 | 0.0575 | 4.4968 | 0.0000 |
| postyes | -4.2746 | -4.7960 | -3.7533 | -16.0709 | 0.0000 |
| corruption | 0.4527 | 0.2280 | 0.6773 | 3.9496 | 0.0001 |

BNEG:

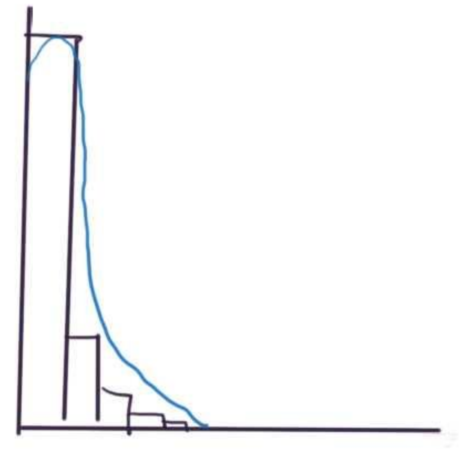
$$\ln(\hat{\lambda}_{bneg_i}) = 1{,}9469 + 0{,}040 \cdot \cancelto{23}{staff_i}$$

$$- 4{,}2746 \cdot \cancelto{0 \;/\; 1}{postyes_i}$$

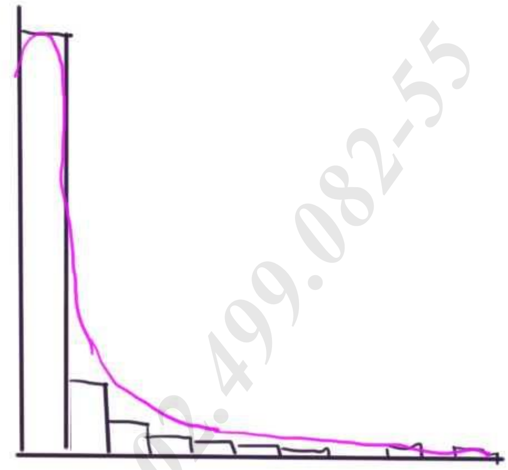
$$+ 0{,}4527 \cdot \cancelto{0{,}5}{corruption_i}$$

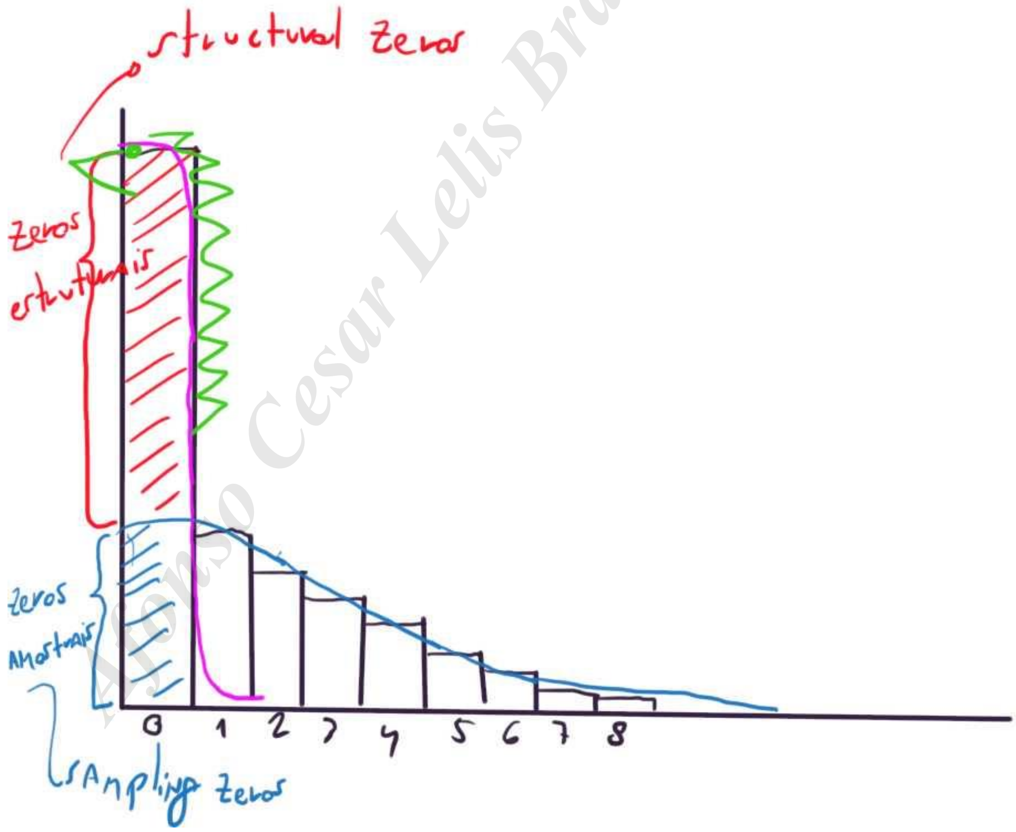


ZIP:



ZINB:




structural zeros
zeros estruturais
zeros amostrais
0 1 2 3 4 5 6 7 8
sampling zeros

```
    Count model coefficients (poisson with log link):
                 Estimate Std. Error z value Pr(>|z|)
α   (Intercept)  2.488857   0.031505  78.999  < 2e-16 ***
β1  corruption   0.093714   0.029982   3.126  0.00177 **
β2  postyes     -4.287651   0.204560 -20.960  < 2e-16 ***
β3  staff        0.020020   0.001239  16.163  < 2e-16 ***

    Zero-inflation model coefficients (binomial with logit link):
                Estimate Std. Error z value Pr(>|z|)
γ   (Intercept)  -1.6117     0.2437  -6.613 3.77e-11 ***
δ1  corruption   -0.9524     0.1955  -4.871 1.11e-06 ***
```

$$\lambda_{ZIP_i} = \left\{1 - \frac{1}{1+e^{-(-1,612 - 0,952 \cdot \cancelto{0,5}{corruption_i})}}\right\} \cdot \left\{e^{(2,489 + 0,020 \times \cancelto{23}{staff_i} \ldots} \; \ldots - 4,288 \cdot \cancelto{0}{postyes_i} + 0,094\, \cancelto{0,5}{corruption_i}\right\}$$

| | | | |
|---|---|---|---|
| GLM | OLS | lm( ) | |
| | Logit | glm(..., family = "binomial") | |
| | mlogit | multinom( ) | → pacote nnet |
| | Poisson | glm(..., family = "poisson") | |
| | bin. negativa | glm.nb( ) | → pacote MASS |
| | ZIP | zeroinfl(..., dist= "poisson") | pacote pscl |
| | ZINB | zeroinfl(..., dist= "negbin") | pacote pscl |